Course outline

About NPTEL

How does an NPTEL online course work?

Week 0

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Week 12

- Research Overview: Higher Order Structures for Graph Explanations
- Research Overview: Representation Surgery
- Summary - Part 1
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- Quiz: Week 12 : Assignment 12
- Week 12 Feedback Form: Responsible & Safe AI Systems
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Week 12 : Assignment 12

The due date for submitting this assignment has passed.

Due on 2024-10-

As per our records you have not submitted this assignment.

1) In graph theory, how many nodes does a single edge connect?

- One node
- Two nodes
- Three nodes
- Any number of nodes

No, the answer is incorrect.
Score: 0
Accepted Answers:
Two nodes

2) Which of the following tasks do Graph Neural Networks (GNNs) typically struggle with?

- Node classification
- Link prediction
- Cycle detection
- Graph clustering

No, the answer is incorrect.
Score: 0
Accepted Answers:
Cycle detection

3) In the context of cell complexes, what does a p-cell represent?

- A cell with p sides
- A cell with p vertices
- An element of dimension p
- A cell with p edges

No, the answer is incorrect.
Score: 0
Accepted Answers:
An element of dimension p

4) What does the acronym FORGE stand for in the context of graph learning?

- Framework for Higher-Optimized Representation in Graph Environments
- Framework for Higher-Order Representations in Graph Explanations
- Functional Optimization for Regular Graph Embeddings
- Fast Operational Research for Graph Equations

No, the answer is incorrect.
Score: 0
Accepted Answers:
Framework for Higher-Order Representations in Graph Explanations

5) After applying FORGE, how do explainers perform compared to Random baselines?

- They consistently surpass Random baselines
- They perform equally to Random baselines
- They occasionally underperform Random baselines
- They consistently underperform compared to Random baselines

No, the answer is incorrect.
Score: 0
Accepted Answers:
They consistently surpass Random baselines

6) Based on the lecture content: What can the boundary relation be loosely translated to in graph theory?

- Nodes
- Edges
- Faces
- Weights

No, the answer is incorrect.
Score: 0
Accepted Answers:

Edges

7) What does guardedness mean as discussed in the lecture?

- Personal information is guarded from being revealed to the outside world due to privacy reasons
- A class is guarded if a classifier can't identify data points belonging to that class
- A model is guarded if you cannot retrieve training data from it
- An attribute is guarded if you can't classify along that attribute

No, the answer is incorrect.
Score: 0
Accepted Answers:
An attribute is guarded if you can't classify along that attribute

8) What is the process/transformation used to achieve guardedness?

- Affine Concept Erasure
- Affine Attribute Erasure
- Affine Model Erasure
- Affine Class Erasure

No, the answer is incorrect.
Score: 0
Accepted Answers:
Affine Concept Erasure

9) How are steering vectors generally defined as discussed in the lecture?

- $v = \mu_0 - \mu_1$ where μ_0 is the mean of undesirable class and μ_1 is the mean of desirable class
- $v = \mu_0 - \mu_1$ where μ_0 is the mean of desirable class and μ_1 is the median of desirable class
- $v = \mu_0 - \mu_1$ where μ_0 is the mean of desirable class and μ_1 is the mean of undesirable class
- $v = \mu_0 - \mu_1$ where μ_0 is the mean of undesirable class and μ_1 is the median of undesirable class

No, the answer is incorrect.
Score: 0
Accepted Answers:
$v = \mu_0 - \mu_1$ where μ_0 is the mean of desirable class and μ_1 is the mean of undesirable class

10) Which of the following is a limitation of graphs as a data structure?

- They can only represent hierarchical relationships
- They can only model pairwise relationships between nodes
- They are restricted to acyclic structures
- They cannot represent directed edges

No, the answer is incorrect.
Score: 0
Accepted Answers:
They can only model pairwise relationships between nodes